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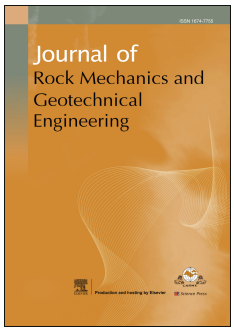

Data- and knowledge-driven three-dimensional geological reconstruction method for tunnel engineering

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Full Length Article

Data- and knowledge-driven three-dimensional geological reconstruction method for tunnel engineering

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Abstract: The creation of a three-dimensional (3D) geological model plays a crucial guiding role in engineering. However, in practice, due to the sparsity of boreholes and the invisibility of strata, accurately reconstructing a 3D geological model has always been a challenging task. In this study, a data- and knowledge-driven 3D geological reconstruction method is proposed, where the Inverse Distance Weighting (IDW) method is integrated with computer vision techniques to improve the accuracy and reliability of geological modeling. The reconstruction of the geological model is realized by the reconstruction of continuous cross-sections in one direction. The reconstruction method integrates two deep learning models: a repair model that learns stratigraphic relationships from borehole data to reconstruct cross-sections, and an interpolation model that predicts intermediate sections by capturing stratigraphic distribution and variation patterns. The comparison with the IDW method and the ordinary kriging method on the virtual data verifies that the proposed method can capture the spatial distribution characteristics of the strata. An engineering example proves that the proposed method can be successfully applied to complex stratum modeling. The proposed method enhances and facilitates intuitive observation of both the reconstructed results and their uncertainties. The proposed method can provide guidance for underground engineering construction sites and contribute to their digital transformation.

Keywords: Three-dimensional geological modeling; Machine learning; Borehole data; Stratigraphic uncertainty; Voxel model

1. Introduction

The safety of tunnel construction highly depends on the geological conditions (Fang et al., 2023a). Creating a reliable geological model is crucial in guiding route selection, construction, and risk assessment. Three-dimensional (3D) geological models have advantages in visualization and digitalization over two-dimensional (2D) drawings (He et al., 2020). However, reconstructing an accurate 3D geological model is a challenging task. Accurate geological information obtained by drilling only accounts for a small part of the construction area, and experts must rely on their knowledge and experience to infer geological information from limited boreholes. This process can lead to subjectivity, resulting in significant uncertainty in the inferred geological information. Therefore, it is essential to establish effective methods for inferring geological information in tunnel engineering.

Geophysical prospecting and drilling are two main methods of obtaining geological data in engineering. Many methods have been proposed for interpreting geological data to obtain stratigraphic distribution information, which can be classified as data-driven and model-driven (Shi and Wang, 2022), such as data-driven interpretation of geophysical information and model-driven interpretation of drilling information. The former acquires internal stratigraphic distribution information with physical detection techniques (such as seismic waves and electromagnetic waves). It then deduces the interface distribution of the geological body by inversion techniques. The latter obtains geological information through geological sketching on tunnel faces, drilling, and other methods. It infers the stratigraphic distribution through modeling methods based on the known geological characteristics. Drilling is one of the most widely applied geological exploration methods in tunnel engineering. Nevertheless, due to project costs, the data obtained through drilling is often sparse, making reliable methods for geological modeling crucial. The currently prevalent methods for geological modeling are based on statistics or probability (Pan et al., 2020). Various statistical methods can be used to predict unknown point attributes. One such method is kriging (Mosammam, 2013), which analyses sampling points based on the variation function. The accuracy of the prediction depends on the skill level of the modeler. Some probability-based methods have been proposed, such as Markov random fields (Li et al., 2016; Wang et al., 2017; Wang et al., 2018) and Markov chains (Zhang et al., 2021; Ouyang et al., 2023). It is essential to determine the transition probability matrix and the direction of the strata for these methods (Li et al., 2019). Additionally, rigorous random field-based methods and conditional random field methods have been proposed for geological modeling (Xie et al., 2023; Yan et al., 2023; Gu et al., 2024). The statistical method relies on the variogram, which only represents the spatial autocorrelation between two points, making it difficult to simulate complex spatial structures (de Vries et al., 2009). To address this issue, the multiple-point statistics (MPS) method has been proposed (Jóhannsson and Hansen, 2021; Zuo et al., 2022). It is a sequential simulation method based on pixels, and training images instead of variogram functions are used to represent the structure of geological spatial. However, it has some limitations, such as its inability to simulate local strata changes that have not appeared in training images (Shi and Wang, 2022) and high computational cost (Zhou et al., 2024). Machine learning methods have become increasingly popular in solving underground engineering problems due to their powerful deep feature mining and learning capabilities (Fang et al., 2023b; Wang et al., 2023; Wang G. et al., 2024). In recent years, several machine learning methods have been proposed to address the challenges associated with

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reconstructing geological models (Zhou et al., 2019; Zhu et al., 2021; Wei and Wang, 2022; Wang H. et al., 2024). Shi and Wang (2021) proposed a 2D geological cross-section prediction method based on the XGBoost model. The algorithm integrates the MPS method with convolutional neural networks to extract stratigraphic connectivity features from training images at different scales. Hou et al. (2023) proposed a 3D geological bodies reconstruction method based on a 2D cross-section by combining the MPS method with a fully connected deep neural network. The approach overcomes the limitations of local optimization in MPS simulations. Zhang et al. (2023) developed a machine learning integration framework, which achieved 3D lithology modeling based on multi-source geophysical data and drilling data, and subdivided the uncertainty in the 3D lithology modeling process through stochastic gradient Langevin enhancement. Nonetheless, these studies that combine machine learning and other methods still have a high computational cost, while pure machine learning methods are limited by the input and output formats of neural networks and require additional post-processing for visual reconstruction. Convolutional neural networks have significant advantages in processing high-dimensional data, extracting spatial features, reducing computational complexity, and generating image output.

This study develops a geological reconstruction method that can reduce computational cost and improve reconstruction efficiency. The method is data- and knowledge-driven and relies on Tobler's first law of geography. Firstly, it reconstructs cross-sectional strata based on borehole data, and the reconstructed parallel 2D cross-section implies the potential anisotropy of the strata. Subsequently, it reconstructs the geological body through continuous cross-sections. The proposed method is validated using both simulated 3D geological bodies and engineering data.

2. 3D geological bodies reconstruction method

In the design and modeling process, borehole data from different locations and sectional information from various directions are often integrated to accurately capture strata anisotropy and reconstruct the 3D model. However, reconstructing precise 3D geological bodies from sparse borehole data remains a significant challenge, as it demands a comprehensive understanding of the spatial variability of strata within the 3D geological bodies. In contrast, reconstructing 2D cross-sections only requires the determination of the spatial variability of strata in the plane. Therefore, reconstructing 2D cross-sections from borehole data is simpler. This study focuses on reconstructing 2D cross-sections that can reveal 3D geological information.

2.1. 3D reconstruction framework based on parallel cross-sections

According to Tobler's first law of geography, cross-sections closer to each other have more similar strata distribution. The strata distribution of the intermediate cross-section can be inferred from the known adjacent parallel cross-sections, enabling the reconstruction of the 3D geological body through continuous cross-section reconstruction. In this process, the known cross-sections contain the distribution characteristics of strata within the cross-sections, and the formation changes between adjacent parallel cross-sections reflect the distribution characteristics of strata in the vertical direction to a certain extent. Therefore, the strata anisotropy is considered in this reconstruction process.

The reconstruction task can be categorized into two types. The first type involves reconstructing the key cross-sections based on borehole data (referred to as the repair task), and the second type involves reconstructing the intermediate cross-sections from the adjacent cross-sections (referred to as the interpolation task). Since the modeling process is linear, the sequence and location of generating key cross-sections play a crucial role in the accuracy of reconstructing the 3D model. In the reconstruction task, more reliable information can help reduce uncertainty in the results. It is important to ensure that each reconstructed key cross-section contains the maximum amount of reliable information. In tunnel engineering, borehole data are the most reliable geological data. Therefore, reconstructing key cross-sections that include the largest number of boreholes is a priority (referred to as the maximum information principle) (Shi and Wang, 2022).

The reconstruction process is shown in Fig. 1, which mainly consists of three aspects: borehole data preprocessing, reconstruction decision-making, and cross-section reconstruction, which will be given specific instructions below.

(1) Drilling data preprocessing: The formatted borehole data is converted into image data through a three-step process comprising data acquisition, standardization, and information pixelization. During the data acquisition phase, each stratum is assigned a unique numerical identifier based on the complete stratigraphic order. In the standardization phase, stratum information from all boreholes is consolidated into a standardized borehole information table, which includes the *X*-coordinate, *Y*-coordinate, top elevation, bottom elevation, and stratum number. Finally, in the information pixelization phase, each stratum is assigned a distinct color with high discriminability, and the borehole information is converted into pixelated representations based on a predefined color table for all strata.

(2) Reconstruction decision-making: Based on borehole coordinates, the reconstruction task (task type and reconstruction location) can be determined. First, the maximum information principle is used to determine the direction of the reconstructed task. As shown in Fig. 1, there are four boreholes in a cross-section along the *X*-axis, but no more than two along the *Y*-axis. According to the maximum information principle, cross-sections parallel to the *Y*-axis should be reconstructed first. Then, based on the reconstruction results, the task type and position are judged. If the cross-section containing the initial borehole is incomplete, a repair task is performed. Otherwise, an interpolation task is carried out. The reconstruction results of the adjacent cross-sections

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are used for the reconstruction of the intermediate cross-sections. Finally, the reconstruction results are evaluated. If the reconstruction does not meet the accuracy requirements, the judgment procedure is triggered to continue cross-section reconstruction. Otherwise, the process ends.

(3) Cross-section reconstruction: The repair task uses the repair model (described in Section 2.3) to reconstruct the cross-section based on the borehole data, and the interpolation task uses the interpolation model (described in Section 2.4) to reconstruct the intermediate cross-section based on the adjacent cross-section.

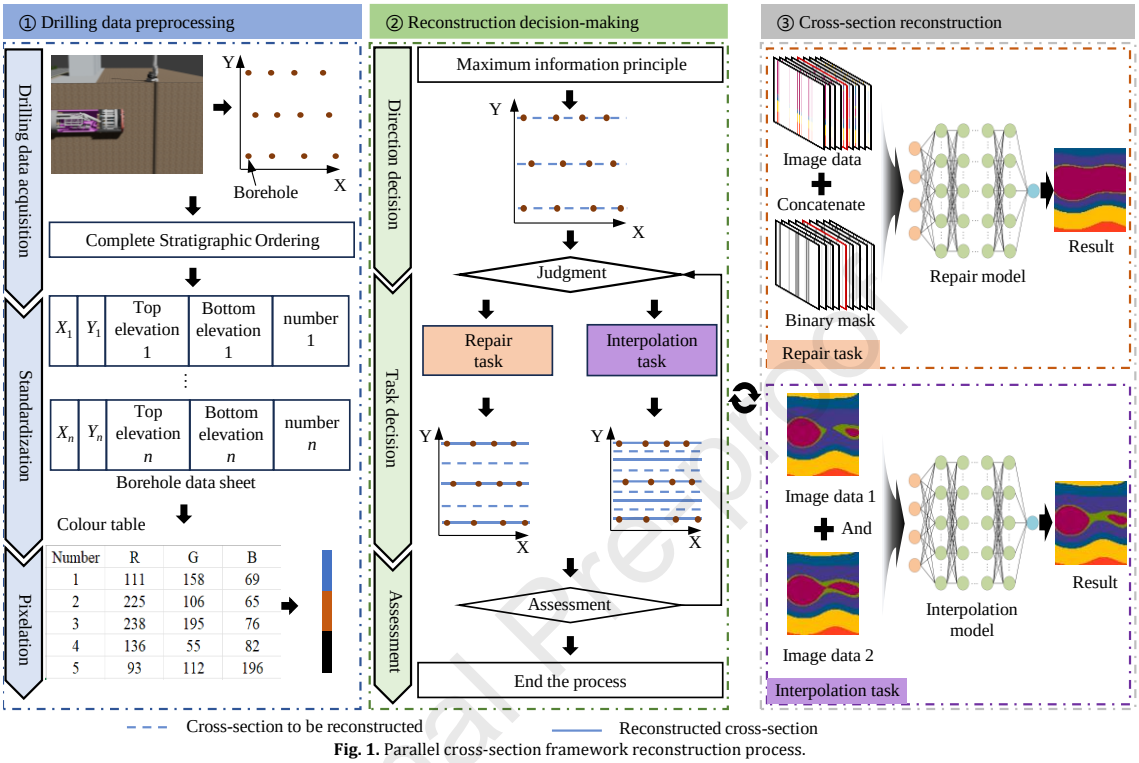

Fig. 1. Parallel cross-section framework reconstruction process.

2.2. Generative adversarial networks

In the field of reconstruction, there are three commonly used methods: variational autoencoder, generative adversarial network (GAN), and diffusion model. Out of the three, GAN has several advantages, such as high flexibility, realistic generation results, and low computational cost. It can generate new data similar to the distribution of data in the training set. GAN consists of a generator network and a discriminator network. The generator aims to produce a plausible output to deceive the discriminator, while the discriminator serves to classify the generated outputs. The objective function of GAN can be expressed as follows:

$$\max_{D}\min_{G} V(G,D) = E_{x_r\sim P_{data}}\left[\ln D(x_r)\right] + E_{x_d\sim P_G}\left[\ln\left(1-D(G(x_d))\right)\right] \quad (1)$$

where $G(x_d)$ is the output of the generator (referred to as x_g in the following), $D(\cdot)$ is the probability that the discriminator determines the input data as real data instead of generated data, and P_{data} and P_G represent the distribution of real data and dirty data, respectively.

The generator produces data with minimal differences from real data, while the discriminator achieves the highest probability of correctly assigning labels. Following the strategy, both networks improve their performance during the training process and generate increasingly accurate outputs. GAN has been used in various image and video applications, such as image denoising (Chen et al., 2018; Kupyn et al., 2019), image restoration (Pathak et al., 2016; Iizuka et al., 2017; Yu et al., 2018), and video frame generation (Wen et al., 2019; Yang and Bors, 2023).

2.3. Repair model for inferring cross-section information from borehole data

The key cross-section can be treated as a damaged image that needs restoration. The borehole data represent the undamaged portions of the image, while the unknown strata distribution represents the damaged areas. The problem of reconstructing key cross-sections can be transformed into an image restoration problem. These cross-sections can use borehole data to deduce information about the unknown strata distribution using image restoration techniques.

The generator and discriminator architecture of the proposed model are shown in Fig. 2. The generator architecture is similar to an autoencoder. The details of its parameters are shown in Table 1. The generator's input consists of two parts: the first is the image to be

repaired (including the reconstructed cross-section and the eight sections before and after its position), and the second is the binary mask corresponding to each image.

Each image is a 3 × 512 × 512 matrix, with the RGB values of the damaged regions set to 255. The corresponding binary mask is a 1 × 512 × 512 matrix, where the value is one at the locations of the damaged areas and zero elsewhere. The two inputs are concatenated into a 9 × 4 × 512 × 512 matrix and then fed into the model. The output of the model is the restored image. The model architecture is inspired by the work of Iizuka et al. (2017). Dilated convolution is used in layers four to seven, which increases the receptive field of the model without increasing the parameters. Compared with standard convolution, this allows the model to utilize further known information to reconstruct images. The model architecture is modified to better meet the task requirements by incorporating a residual network architecture at the locations of dilated convolutions. The architecture of the residual network is shown in Fig. 3. The residual block can be represented as:

$$\hat{x} = x + f(x, \omega) \tag{2}$$

where x and $\hat{x}$ represent the input and output of the residual block, respectively; ω represents the weight of the residual block; and $f(\cdot)$ represents the fitting function.

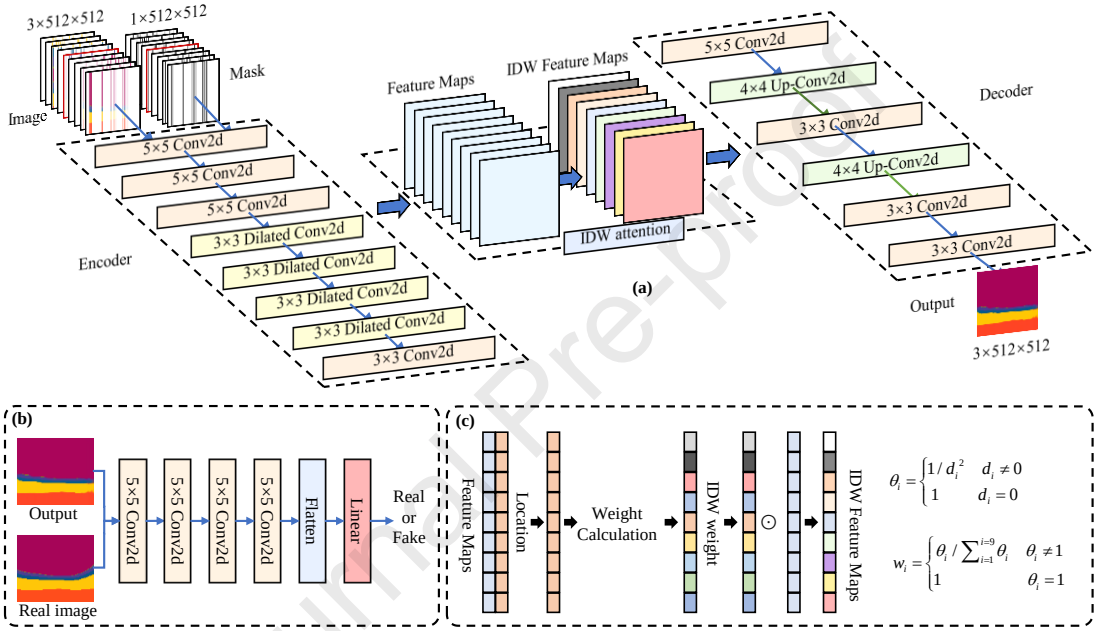

Fig. 2. Section reconstruction model architecture: (a) architecture of generator, (b) architecture of discriminator, and (c) architecture of IDW attention.

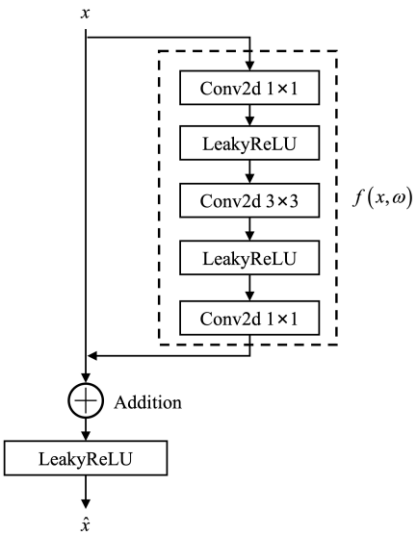

Fig. 3. Residual network architecture.

The residual network consists of two parts: the direct mapping structure and the residual structure. The left side of Fig. 3 shows the direct mapping structure, and the right shows the residual structure. The direct mapping structure addresses the degradation problem associated

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with deep models. The residual structure is composed of three convolutional networks, and the LeakyReLU is used as the activation function. The 1 × 1 convolution kernels in the residual structure help reduce the parameters of the network and training costs.

The IDW attention is shown in Fig. 2c. d_i represents the distance from the i-th cross-section to the reconstructed cross-section, $\underline{\theta_i}$ represents the weight of the i-th cross-section, and w_i is the normalized weight of the i-th cross-section. It assigns weights to each reference section by calculating the distance between the reference sections and the reconstructed section. The reference section refers to the eight sections before and after the reconstructed cross-section in the generator's input. If the number of reference sections is insufficient, a full zero matrix is generated instead, with an IDW weight of zero.

Table 1
Structure of the generator.

Layer ID	Detail	Output size
1	Conv2d (kernel_size=5, stride=1, padding=2) LeakyReLU	9 × 32 × 512 × 512
2	Conv2d (kernel_size=5, stride=2, padding=2) LeakyReLU	9 × 64 × 256 × 256
3	Conv2d (kernel_size=5, stride=2, padding=2) LeakyReLU	9 × 256 × 128 × 128
4	Residual Block (kernel_size=3, stride=1, padding=2, dilation=2) LeakyReLU	9 × 256 × 128 × 128
5	Residual Block (kernel_size=3, stride=1, padding=4, dilation=4) LeakyReLU	9 × 256 × 128 × 128
6	Residual Block (kernel_size=3, stride=1, padding=8, dilation=8) LeakyReLU	9 × 256 × 128 × 128
7	Residual Block (kernel_size=3, stride=1, padding=16, dilation=16) LeakyReLU	9 × 256 × 128 × 128
8	Conv2d (kernel_size=3, stride=1, padding=1) LeakyReLU	9 × 128 × 128 × 128
9	Conv2d (kernel_size=5, stride=1, padding=2) LeakyReLU	1 × 128 × 128 × 128
10	ConvTranspose2d (kernel_size=4, stride=2, padding=1) LeakyReLU	1 × 64 × 256 × 256
11	Conv2d (kernel_size=3, stride=1, padding=1) LeakyReLU	1 × 64 × 256 × 256
12	ConvTranspose2d (kernel_size=4, stride=2, padding=1) LeakyReLU	1 × 64 × 512 × 512
13	Conv2d (kernel_size=3, stride=1, padding=1) LeakyReLU	1 × 32 × 512 × 512
14	Conv2d (kernel_size=3, stride=1, padding=1) Sigmoid	1 × 3 × 512 × 512

The discriminator comprises four convolutional layers and one fully connected layer. The details of its parameters are shown in Table 2. It receives input data as a 3 × 512 × 512 matrix and outputs a probability value between zero and one, indicating the likelihood of the input image being real.

Table 2
Structure of the discriminator.

Layer ID	Detail	Output size
1	Conv2d (kernel_size=5, stride=2, padding=2) LeakyReLU	64 × 256 × 256
2	Conv2d (kernel_size=5, stride=2, padding=2) LeakyReLU	128 × 128 × 128
3	Conv2d (kernel_size=5, stride=2, padding=2) LeakyReLU	256 × 64 × 64
4	Conv2d (kernel_size=5, stride=2, padding=2) LeakyReLU	512 × 32 × 32
5	Flatten	524,288
6	Linear Sigmoid	1

2.4. Interpolation model for inferring 3D volume information from sectional data

The key cross-sections reconstructed in Section 2.3 not only show the spatial distribution information of the geological body in these sections but also imply the spatial distribution characteristics in the direction perpendicular to these sections through strata changes between sections. By learning from the strata distribution characteristics of these continuous sections, the neural network can understand the spatial variability of the geological bodies.

The problem of video interpolation requires analyzing consecutive image frames to identify patterns of object motion and changes within the images. This information is then used to infer the distribution of images in the intermediate frames. Similarly, in the sectional interpolation problem, the adjacent key cross-sections are analyzed to determine the patterns of geological changes. Based on these patterns, the distribution of geological layers in the intermediate cross-sections can be inferred. Therefore, the problem of reconstructing geological bodies by continuous parallel 2D sections can be transformed into a video interpolation problem. By using the video interpolation algorithm, non-linear movements can be captured, which also helps in identifying irregular variations in geological layers.

The interpolation model is established by referring to the U-Net network architecture, which consists of an encoder, a feature fusion layer, a self-attention layer that employs the Squeeze-and-Excitation Network (SENet) (Hu et al., 2018), and a decoder. The details of its parameters are shown in Table 3. The shape of the model's input and output is identical, which is a 3 × 512 × 512 matrix. The model architecture is shown in Fig. 4a. The encoder is responsible for extracting features from two images. The feature fusion layer concatenates the feature maps of both images along the channel dimension. It reduces the dimensionality using 1 × 1 convolution to maintain consistency in the output and input data dimensions. The architecture of the self-attention layer is shown in Fig. 4b. In the self-attention layer, the original data is transformed into a 1 × 1 × C linear vector at first, which passes through an activation layer, assigning weights between zero and one to each channel. Then, this

weight vector is multiplied by the original matrix to weigh each channel, enabling the network to learn features better for the reconstruction task. The processing flow of the feature fusion layer and the self-attention layer is shown in Fig. 4c. The decoder is responsible for transforming the fused lower-level features back into a reconstructed image through multiple deconvolutions.

Table 3
Interpolation model structure.

Layer ID	Detail	Output size
1	Residual Block (kernel_size=4, stride=2, padding=1) LeakyReLU	64 × 256 × 256
2	Residual Block (kernel_size=4, stride=2, padding=1) LeakyReLU	128 × 128 × 128
3	Residual Block (kernel_size=4, stride=2, padding=1) LeakyReLU	256 × 64 × 64
4	Residual Block (kernel_size=4, stride=2, padding=1) LeakyReLU	512 × 32 × 32
5	Residual Block (kernel_size=4, stride=2, padding=1) LeakyReLU	1024 × 16 × 16
6	SENet	1024 × 16 × 16
7	ConvTranspose2d (kernel_size=4, stride=2, padding=1) LeakyReLU	512 × 32 × 32
8	ConvTranspose2d (kernel_size=4, stride=2, padding=1) LeakyReLU	256 × 64 × 64
9	ConvTranspose2d (kernel_size=4, stride=2, padding=1) LeakyReLU	128 × 128 × 128
10	ConvTranspose2d (kernel_size=4, stride=2, padding=1) LeakyReLU	64 × 256 × 256
11	ConvTranspose2d (kernel_size=4, stride=2, padding=1) Sigmoid	3 × 512 × 512

The U-Net model has been modified to improve its performance in image-related tasks. The modifications are as follows:

(1) The residual network is used to reduce computational cost.

(2) The Batch Normalization layer after the convolutional layers was removed, as experiments showed it accelerates training but causes color deviation artifacts in generated images for image-related tasks.

(3) The model extracts features from the bottom-most layers of the images, and after fusion, a channel attention layer is established to select the most beneficial features for the reconstruction task.

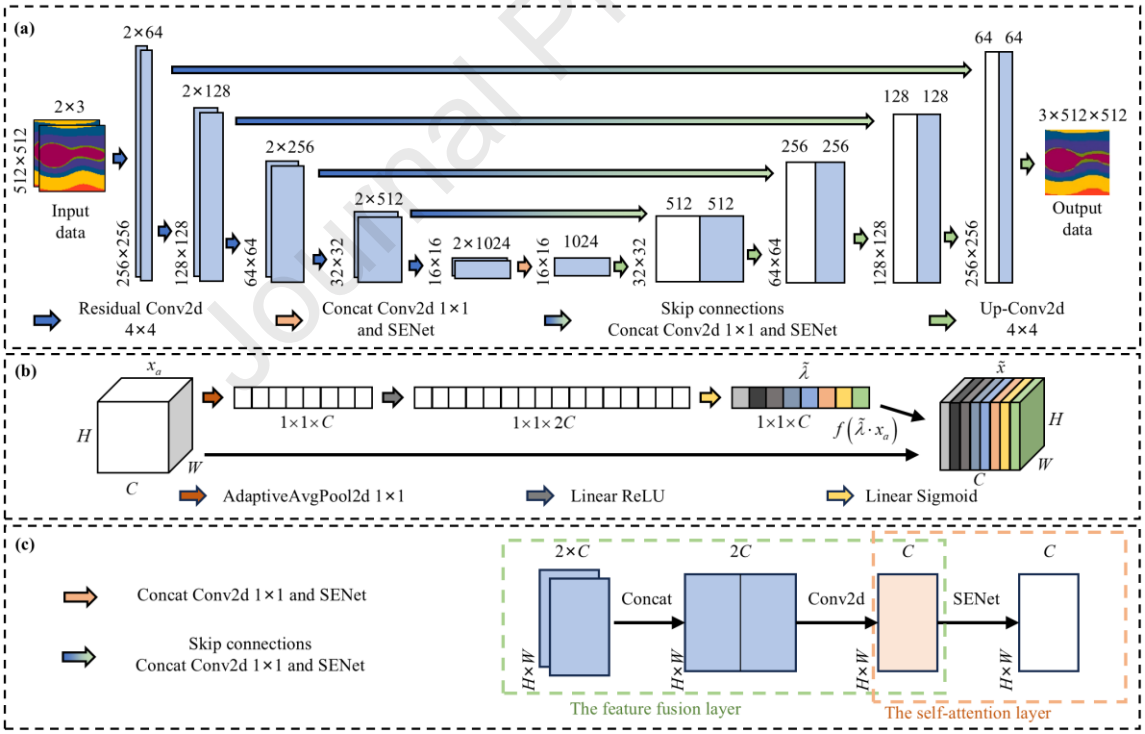

Fig. 4. Proposed architecture for interpolation model. (a) proposed architecture for interpolation model, (b) architecture of self-attention layer, and (c) concat convolution and self-attention process.

In Fig. 4c, x_a represents the input data of the self-attention layer, C represents the number of feature channels for x_a, H represents the feature height of x_a, W represents the feature width of x_a, $\tilde{\lambda}$ represents the weight value of each channel, and $\tilde{x}$ represents the output of the self-attention layer.

2.5. Loss function

The training process of GAN relies on a loss function that measures the difference between the generated data and the real data (Goodfellow et al., 2014). In the cross-section reconstruction model, two types of loss functions work together to evaluate the similarity of generated images: adversarial loss and reconstruction loss. In the volumetric reconstruction model, only the reconstruction loss is considered. The overall loss function is as follows:

$$Loss = L_{\text{adv}}\left(y,\hat{y}\right) + L_{1}\left(x_{\text{r}},x_{\text{g}}\right) + L_{\text{GDL}}\left(x_{\text{r}},x_{\text{g}}\right) + L_{\text{NPCL}}\left(x_{\text{r}},x_{\text{g}}\right) \quad (3)$$

where y represents the label values (the real image is one and the generated image is zero), $\hat{y}$ represents the discrimination values generated by the discriminator (the values range from zero to one). L_{adv} can be expressed as a binary cross-entropy function.

$$L_{\text{adv}}\left(y,\hat{y}\right) = -\left(y\ln\left(\hat{y}\right) + \left(1-y\right)\ln\left(1-\hat{y}\right)\right) \quad (4)$$

The remaining three losses in the overall loss function are reconstruction loss, which is used for training the generator, where MAE is as follows:

$$L_{1}\left(x_{\text{r}},x_{\text{g}}\right) = \frac{1}{n}\sum_{j=1}^{n}\left|\left(x_{\text{r}}\right)_{j} - \left(x_{\text{g}}\right)_{j}\right| \quad (5)$$

where n represents the number of pixels in an image.

Mean absolute error (MAE) and mean squared error (MSE) are the popular per-pixel comparison methods, which constrain the distribution of the output image as close as possible to the real image. Compared to MSE, MAE loss helps in sharpening the output image (Zhao et al., 2017).

The gradient difference loss (GDL) considers the relative differences between adjacent pixels in x_{g} and x_{r}.

$$L_{\text{GDL}}\left(x_{\text{r}},x_{\text{g}}\right) = \sum_{j,k}\left(\left\|\left|\left(x_{\text{r}}\right)_{j,k} - \left(x_{\text{r}}\right)_{j-1,k}\right| - \left|\left(x_{\text{g}}\right)_{j,k} - \left(x_{\text{g}}\right)_{j-1,k}\right|\right\|^{\alpha} + \left\|\left|\left(x_{\text{r}}\right)_{j,k} - \left(x_{\text{r}}\right)_{j,k-1}\right| - \left|\left(x_{\text{g}}\right)_{j,k} - \left(x_{\text{g}}\right)_{j,k-1}\right|\right\|^{\alpha}\right) \quad (6)$$

where α is an integer greater than or equal to one and is set as one in this research. $j \in H_{\text{image}}$, $k \in W_{\text{image}}$.

The normalized product correlation loss (NPCL) is also incorporated into the loss function, which is a commonly used metric in image matching.

$$R\left(x_{\text{r}},x_{\text{g}}\right) = \sum_{j,k}\left[\left(x_{\text{r}}\right)_{j,k} \times \left(x_{\text{g}}\right)_{j,k}\right] \Big/ \sqrt{\sum_{j,k}\left(x_{\text{r}}\right)_{j,k}^{2} \times \sum_{j,k}\left(x_{\text{g}}\right)_{j,k}^{2}} \quad (7)$$

$$L_{\text{NPCL}}\left(x_{\text{r}},x_{g}\right) = -\ln\left(R\left(x_{\text{r}},x_{g}\right)\right) \quad (8)$$

$j \in H_{\text{image}}$, $k \in W_{\text{image}}$. The logarithmic operation converts its value range from [0, 1] to (∞, 0]. When $R(x_{\text{r}},x_{\text{g}})$ equals one and $L_{\text{NPCL}}(x_{\text{r}},x_{\text{g}})$ is zero, it can be found that $L_{\text{NPCL}}(x_{\text{r}},x_{\text{g}})$ is a convex function. Compared to MSE and GDL, NPCL is scale-free and resilient to white noise. Therefore, it measures the difference between real frames and generated frames from another perspective.

3. Methods for training and evaluating models

3.1. Training data

The training data is derived from Gempy, a free and open-source Python library. Gempy (de la Varga et al., 2019) is an implicit 3D geological modeling software that allows the creation of complex geological models, such as folds, faults, and unconformities. The fundamental principle behind Gempy is the theory of potential fields. It considers uncertainties in modeling, implements stochastic geological modeling, and achieves Bayesian inversion. The implicit modeling capabilities of Gempy have been used to assist scientific research (Fandel et al., 2021; Jessell et al., 2021; Thomas et al., 2022). In this research, random geological bodies are generated using Gempy, and random borehole data is extracted from these bodies to train and test the deep learning model. A database consisting of 18,466 images was generated by randomly creating 50 geological bodies with *XYZ* coordinates ranging between 600 and 800 units in length. The number of soil types for these bodies is a random number between 3 and 8. It encompasses geological features, such as folds, interbedding, and lenses in its strata distribution. These bodies are sliced at intervals of 2 units. Some training images are shown in Fig. 5. Each row of images represents cross-sections of the same geological body at different locations. Since the purpose of the models is to learn the characteristics of strata distribution under different geological formations, the images do not contain any attribute information. In the repair task, some binary masks are randomly generated with the same size as the image. These masks are multiplied with the image to create the damaged part of the image. For the interpolation task, its database is derived from the original database, with a total of 9210 samples in the database. Each sample consists of three images of the same geological body arranged by their spatial position, where the first and last images in the sequence are input data, and the intermediate image acts as the label data. To effectively learn spatial relationships across different scales, the three images are not always immediately adjacent in space. For each geological body, the generated data were split into training and test sets in a ratio of 7:3.

3.2. Interpolation methods for comparison

The IDW and ordinary kriging are two widely used deterministic interpolation methods in geological reconstruction tasks (Liu et al., 2020). Their high computational efficiency and flexible adjustment of parameters make them suitable for comparing reconstruction effects in contrast experiments. In IDW, the value of a prediction point is estimated by considering the values of neighboring sample points. The method assumes that sample points closer to the prediction point have more influence on the results and that the stratigraphic distribution is isotropic. However, the strata distribution is often anisotropic in practical engineering, and the anisotropy ratio is used to describe the spatial anisotropy of a formation. The estimated value ($\hat{Z}_0$) can be calculated using the following formulas.

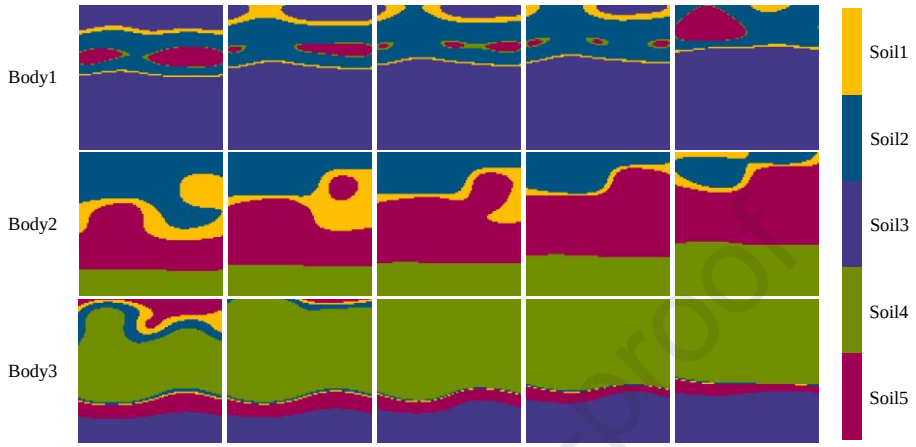

Fig. 5. Some training images of the training database.

$$\hat{Z}_0 = \sum_{i=1}^{m} w_i Z_i \tag{9}$$

$$w_i = \frac{1/d_i^{\beta}}{\sum_i^m 1/d_i^{\beta}} \tag{10}$$

$$d_i = \sqrt{(x_i - x_0)^2 + \lambda (y_i - y_0)^2} \tag{11}$$

where Z_i represents the value of sampling point i; (x_i, y_i) and (x_0, y_0) represent the Cartesian coordinates of sampling point i and predicted point; m represents the number of all sampling points; w_i represents the weight of sampling point i; d_i represents the spatial distance between the sampling point i and the predicted point, with β typically taken as one or two. When λ is equal to one, the spatial distribution is considered to be isotropic. When λ is greater than one, spatial proximity weakens in the Y-direction. Conversely, when λ is less than one, spatial proximity weakens in the X-direction. This method is straightforward to compute and user-friendly, yielding satisfactory reconstruction results when sampling points are relatively uniformly distributed. However, it has some drawbacks. First, the parameter β is not predetermined and requires users to determine it based on experience. Then, the reciprocal function cannot describe spatial correlation precisely.

Kriging methods were proposed to address these issues. The prediction formula of ordinary kriging is also based on weighted calculation. The difference is that the weights used in ordinary kriging are not simply the reciprocals of distances between points but a set of optimal coefficients that minimize the difference between the estimated value and the actual value Z_0.

$$\min_{\lambda_i} Var\left(\hat{Z}_0 - Z_0\right) \tag{12}$$

It simultaneously fulfils the condition of an unbiased estimation.

$$E\left(\hat{Z}_0 - Z_0\right) = 0 \tag{13}$$

3.3. Quantitative evaluation methods

To compare different reconstruction methods, three functions are selected to describe the difference between the reconstructed strata and the true strata. These functions are global accuracy (GA), MSE, and multi-scale structural similarity (MS-SSIM).

The formula for global accuracy is as follows:

$$GA = \frac{\sum_{i=1}^{n} I\left(T_{pi} = T_{ai}\right)}{n} \tag{14}$$

where T_{pi} represents the predicted strata type of point i; T_{ai} represents the actual strata type of point i; I represents the indicator function, and has a value of one when the predicted result equals the actual strata type; otherwise, it is zero.

The formula for MSE is as follows:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}\left(\left(x_{\mathrm{g}}\right)_i - \left(x_{\mathrm{r}}\right)_i\right)^2 \tag{15}$$

$$SSIM\left(x_{\mathrm{r}}, x_{\mathrm{g}}\right) = \left[l\left(x_{\mathrm{r}}, x_{\mathrm{g}}\right)^{\varphi} \cdot c\left(x_{\mathrm{r}}, x_{\mathrm{g}}\right)^{\chi} \cdot s\left(x_{\mathrm{r}}, x_{\mathrm{g}}\right)^{\gamma}\right] = \frac{\left(2\mu_{x_{\mathrm{r}}}\mu_{x_{\mathrm{g}}} + C_1\right)\left(2\sigma_{x_{\mathrm{r}}x_{\mathrm{g}}} + C_2\right)}{\left(\mu_{x_{\mathrm{r}}}^2 + \mu_{x_{\mathrm{g}}}^2 + C_1\right)\left(\sigma_{x_{\mathrm{r}}}^2 + \sigma_{x_{\mathrm{g}}}^2 + C_2\right)} \tag{16}$$

$$\mu_{x_{\mathrm{g}}} = \frac{1}{n}\sum_{i=1}^{n}\left(x_{\mathrm{g}}\right)_i \tag{17}$$

$$\mu_{x_{\mathrm{r}}} = \frac{1}{n}\sum_{i=1}^{n}\left(x_{\mathrm{r}}\right)_i \tag{18}$$

$$\sigma_{x_{\mathrm{g}}} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}\left(\left(x_{\mathrm{g}}\right)_i - \mu_{x_{\mathrm{g}}}\right)^2} \tag{19}$$

$$\sigma_{x_{\mathrm{r}}} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}\left(\left(x_{\mathrm{r}}\right)_i - \mu_{x_{\mathrm{r}}}\right)^2} \tag{20}$$

$$\sigma_{x_{\mathrm{r}}x_{\mathrm{g}}} = \frac{1}{n-1}\sum_{i=1}^{n}\left(\left(x_{\mathrm{g}}\right)_i - \mu_{x_{\mathrm{g}}}\right)\left(\left(x_{\mathrm{r}}\right)_i - \mu_{x_{\mathrm{r}}}\right) \tag{21}$$

$$C_1 = \left(K_1 L\right)^2 \tag{22}$$

$$C_2 = \left(K_2 L\right)^2 \tag{23}$$

$$L = 2^{\mathrm{bit}} - 1 \tag{24}$$

where l, c, and s represent the equations for calculating brightness, contrast, and structural similarity, respectively; and n represents the number of pixels in an image; K_1 and K_2 are set to be 0.01 and 0.03 by default, respectively; L is the dynamic range of pixels, which is typically taken to be 255 for 8-bit images; C_1 and C_2 are used to prevent the denominator from being zero and maintain the stability of the calculation. Once K_1, K_2, and L are determined, C_1 and C_2 become fixed constants.

In practical applications, evaluating the quality of an image is often done by analyzing a set of windows instead of the entire image. The perceptibility of image details relies on the sampling density of the image signal, the distance from the image plane to the observer, and the perceptual capabilities of the visual system of the observer (Wang et al., 2004). Therefore, the multiscale structural similarity is commonly used to capture image details at different resolutions. The range of values for MS-SSIM is [0, 1], where a larger value indicates a smaller difference between the two images being compared.

$$MS-SSIM\left(x_{\mathrm{r}}, x_{\mathrm{g}}\right) = \left[l\left(x_{\mathrm{r}}, x_{\mathrm{g}}\right)\right]^{\alpha_M} \cdot \prod_{j=1}^{M}\left[c\left(x_{\mathrm{r}}, x_{\mathrm{g}}\right)\right]^{\beta_j} \cdot \left[s\left(x_{\mathrm{r}}, x_{\mathrm{g}}\right)\right]^{\gamma_j} \tag{25}$$

where α_M, β_j, and γ_j are used to adjust the relative importance of different components, and M is the number of scales. Wang et al. (2003) determined the values of these parameters when the number of scales was five through experiments. These values have since been widely adopted in subsequent studies. The specific values of the parameters are as follows:

$$\beta_1 = \gamma_1 = 0.0448 \tag{26}$$

$$\beta_2 = \gamma_2 = 0.2856 \tag{27}$$

$$\beta_3 = \gamma_3 = 0.3001 \tag{28}$$

$$\beta_4 = \gamma_4 = 0.2363 \tag{29}$$

$$\alpha_5 = \beta_5 = \gamma_5 = 0.1333 \tag{30}$$

3.4. Uncertainty quantification

In this section, we discuss the quantification of uncertainty in strata reconstruction. The uncertainty of the reconstruction result is quantified by information entropy. Information entropy was proposed by Shannon (1948) to reflect the average uncertainty of all information that may be generated by an information source. Information entropy has been widely used to characterize uncertainty in geology (Yan et al., 2023; Hu et al., 2024). The information entropy at a specific position i can be expressed as

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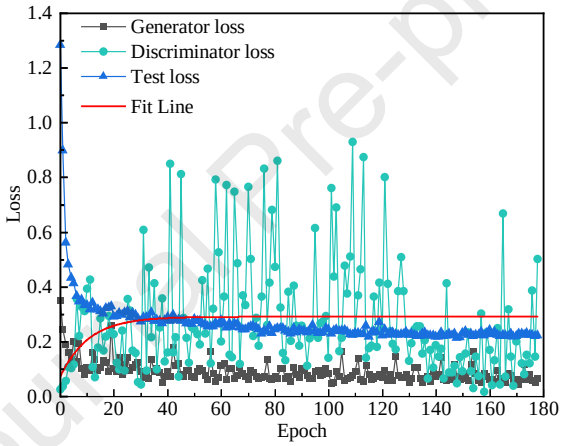
$$H(i) = -\sum_{s=1}^{k} p_s(i) \ln p_s(i) \qquad (31)$$

where k represents the number of stratum types in the reconstructed section, and $p_s(i)$ represents the probability of soil s at position i. The probability will be estimated by a large number of experiments based on the law of large numbers. The greater the entropy, the greater the uncertainty of the location i based on existing information.

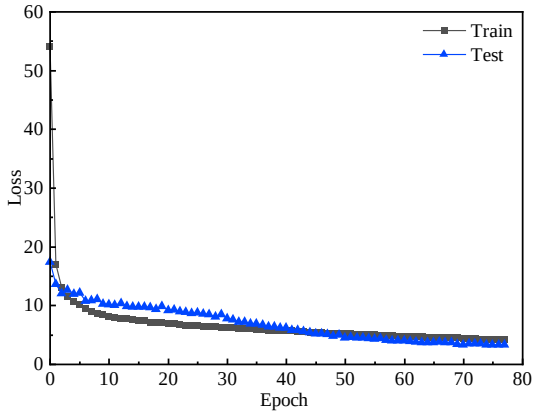
Experts may have divergent opinions on geological modeling results based on their distinct knowledge and experience. A neural network encodes its learned knowledge in its neuronal parameters. Dropout randomly deactivates subsets of neurons during inference, effectively simulating different experts by varying the model's information pathways. By statistically analyzing multiple dropout-induced reconstructions, we can estimate the uncertainty of the predicted stratum. To facilitate uncertainty estimation, dropout layers are inserted after each layer except the output layer in this study.

3.5. Model training

Fig. 6 illustrates the changes in evaluation metrics for both the repair and interpolation models on the training and test sets. In Fig. 6a, the repair model's generator and test losses drop sharply during the first 20 iterations, indicating that the model efficiently learns stratum distribution features at the early stages of training. The discriminator loss is calculated using binary cross-entropy, with its ideal value being −ln(0.5) (approximately 0.693), indicating that the discriminator cannot distinguish real from generated images and outputs 0.5 probability for both. The abnormally low loss indicates that the generator lacks sufficient capacity. In this study, the loss converged at approximately 0.3. Both generator and test losses continue to decline, stabilizing around 50 iterations, which indicates model convergence. For the interpolation model (Fig. 6b), loss falls sharply in the first 5 iterations and continues to decrease until about 40 iterations, after which it converges.



(a) Repair model



(b) Interpolation model

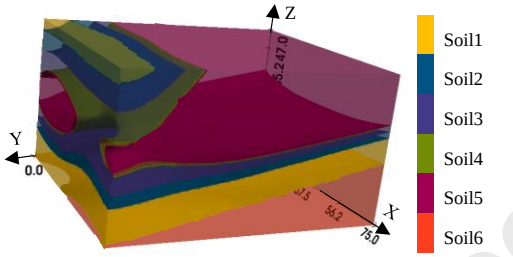
Fig. 6. Relationship between evaluation indicators and iteration rounds of models.

4. Material and methods

In this section, comparative experiments are conducted to evaluate the reconstruction effect of the proposed method. The database introduced in Section 3.1 is used to train networks. The reconstruction experiments are performed using the same simulated geological body.

4.1. Simulated geological body

Gempy was used to generate the random geological body with dimensions of 75 m (*X*-distance) × 86 m (*Y*-distance) × 47 m (*Z*-distance). The geological body consists of six different geological layers, and their specific distribution is shown in Fig. 7. Some data were extracted to simulate the borehole data. These boreholes are distributed on 18 sections perpendicular to the *Y*-axis, and the interval between these sections is five meters. The distribution of randomly generated borehole data on each cross-section is shown in Fig. 8. The boreholes are intentionally randomized rather than evenly spaced for two primary reasons: (1) Boreholes are not always uniformly spaced in practical engineering scenarios. (2) Random boreholes may overlap, resulting in the repetition of some geological information. This overlapping phenomenon results in different proportions of known information in different cross-sections, and it is used to demonstrate the model's generalization capability.



(a) The 3D geological body

(b) Interpolation model, Geological distribution of cross-section at X = 75

Fig. 7. Strata distribution of the geological body.

4.2. Cross-section reconstruction task

Five cross-sections with *Y*-coordinates of 0, 10, 15, 20, and 85 from the geological body in Section 4.1 are selected as test sections. These cross-sections are discretized into 512 × 512 grids. Ten nearest sampling points are searched to calculate the attribute of the prediction point in the IDW, and sixteen nearest sampling points are searched to calculate the attribute of the prediction point in the ordinary kriging. The cross-sectional reconstruction results of the three methods are shown in Fig. 9, and the quantitative losses are shown in Table 4. Since the colors in both the real image and the predicted image use the same color scheme, both images are converted to grayscale. The similarity between them is measured by the difference value, which serves as the loss. The differences between the results and the real data are shown in the loss cloud maps in Fig. 10. The borehole position is represented by a white dotted line. Notably, the jagged distribution of interfaces is evident in the reconstruction results of the IDW and ordinary kriging methods. This is because the two methods consider only some adjacent points for interpolation. When the strata of adjacent boreholes change significantly, the interpolation results are not smooth due to sudden changes in sampling points. This is also why the mutations of the strata occur near the middle of the adjacent boreholes. This phenomenon can be effectively addressed by the proposed method. The proposed method outperforms both IDW and ordinary Kriging across the four quantitative indicators. Geological reconstruction tasks are challenging, particularly in predicting the boundaries of the geological body. The three methods show significant errors at the boundaries of the geological body in the loss maps. However, the proposed method consistently exhibits reduced errors at the majority of geological body boundaries in comparison to the other two methods. Therefore, the proposed method can infer the spatial distribution characteristics of strata in the cross-section based on the borehole data.

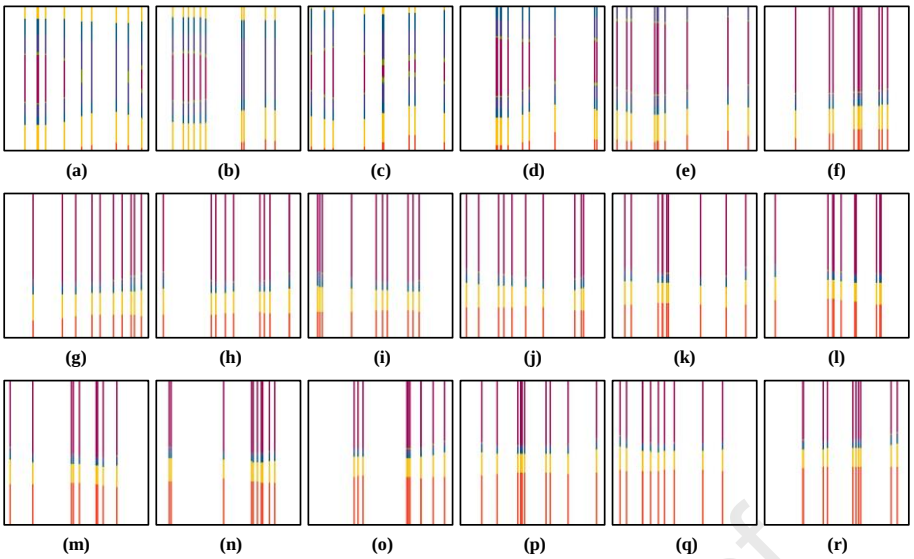


Fig. 8. Distribution of randomly generated borehole data on each cross-section. (a) Y = 0 m, (b) Y = 5 m, (c) Y = 10 m, (d) Y = 15 m, (e) Y = 20 m, (f) Y = 25 m, (g) Y = 30 m, (h) Y = 35 m, (i) Y = 40 m, (j) Y = 45 m, (k) Y = 50 m, (l) Y = 55 m, (m) Y = 60 m, (n) Y = 65 m, (o) Y = 70 m, (p) Y = 75 m, (q) Y = 80 m, and (r) Y = 85 m.

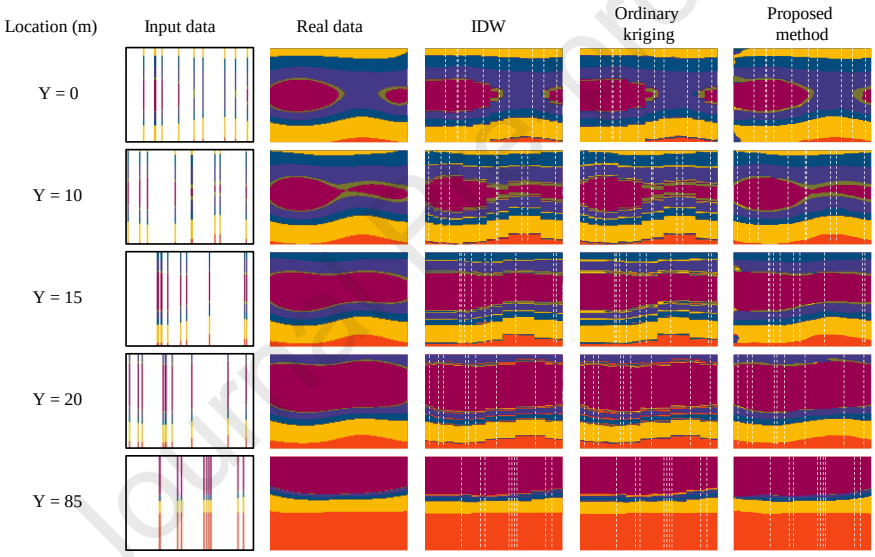


Fig. 9. Reconstruction performance of three reconstruction methods on selected cross-sections.

Table 4

Reconstruction losses of the three reconstruction methods on selected cross-sections.

Index	IDW	Ordinary kriging	Proposed method
GA	0.9282	0.9206	**0.9364**
MSE	686.0483	818.3088	**519.7413**
SSIM	0.8828	0.8736	**0.92**
MS-SSIM	0.841	0.8184	**0.8823**

A total of 1000 reconstruction experiments were performed using the method proposed in Section 3.4. The distribution of the most likely strata and the information entropy of the strata are shown in Fig. 11. According to the information entropy distribution, the uncertainty of the stratum is mainly concentrated at the boundary of the stratum. When the stratum between the two boreholes changes significantly, it leads to uncertainty in the stratum distribution. Large spacing of boreholes will also lead to uncertainty in the strata distribution.

To assess the effect of training data volume on restoration model performance, six datasets were constructed of varying sizes, containing 5%, 20%, 40%, 60%, 80%, and 100% of the total training data, respectively. Eighteen cross-sections (Fig. 8) and a complete test set were reserved for testing. The test results for these six models are presented in Fig. 12. Fig. 12a illustrates that increasing the training data proportion from 5% (≈646 samples) to 100% (≈12,926 samples) does not significantly alter the model's performance on new strata. The observed fluctuations primarily stem from the intrinsic instability of GAN training. In Fig. 12b, when the data volume increased from 5% to 40%, the model showed a relatively larger improvement in performance, but the absolute performance gain remained small. Therefore, model

performance depends on both data diversity and dataset size, and once the dataset reaches a sufficient scale, diversity becomes the dominant factor. The training dataset should encompass a broad spectrum of stratigraphic distribution features.

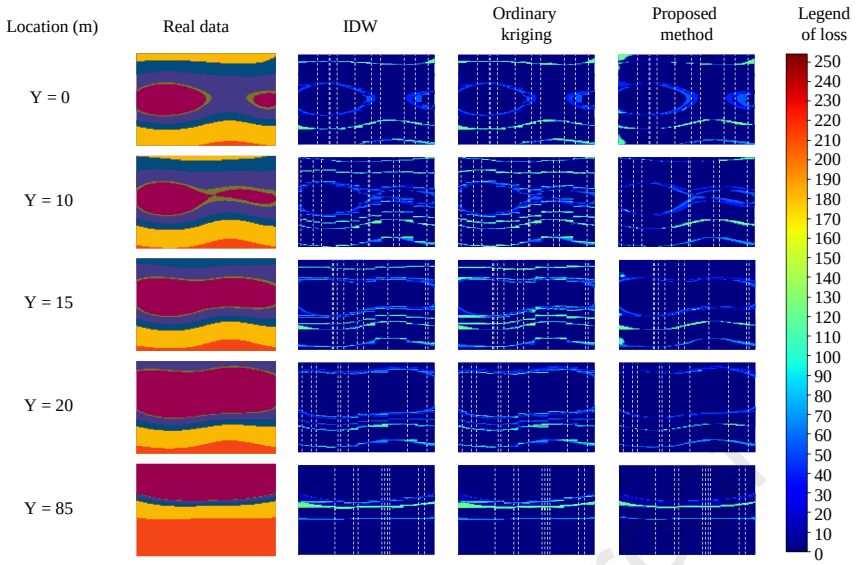


Fig. 10. Loss maps of three reconstruction methods on selected cross-sections.

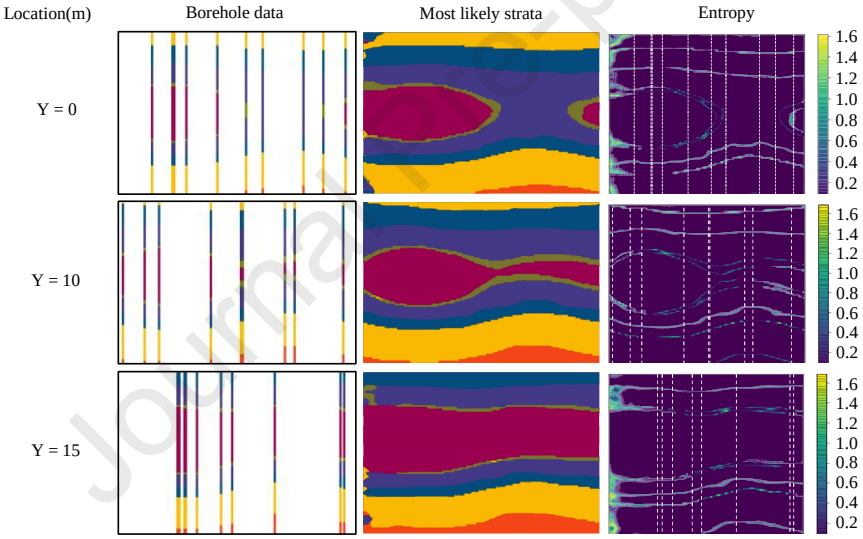
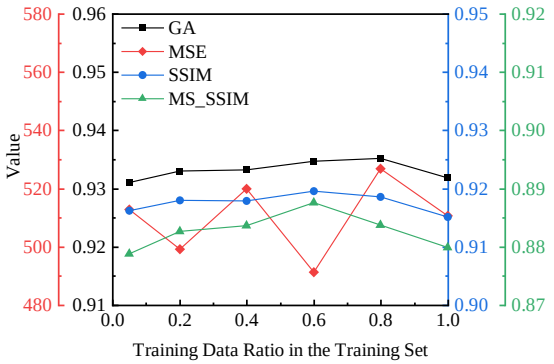
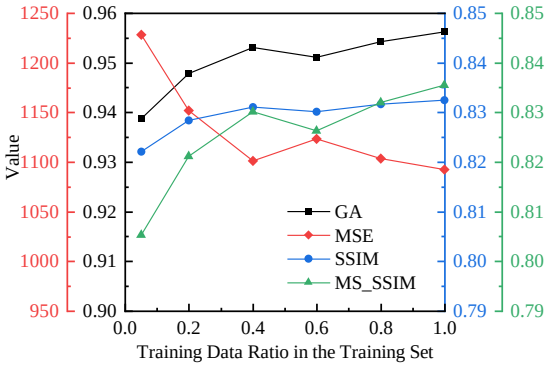


Fig. 11. Uncertainty of proposed methods on selected cross-sections.



(a) test on 18 cross-sections



(b) test on test set

Fig. 12. Impact of training data volume.

4.3. Geological body reconstruction task

In the volumetric reconstruction task, the attributes of the prediction points are computed using the same number of sampled points, as shown in Section 4.2. The reconstruction results of the three methods are shown in Table 5 and Fig. 13. As shown in Table 5, the proposed method is better than the other two methods in four quantitative indicators. In the voxel models reconstructed by IDW and ordinary kriging, there are irregular fluctuations at the upper layer boundaries on the surface. This does not match the actual distribution of the geological layers. Conversely, the geological body reconstructed by the proposed method is more similar to the real stratigraphy. The results show that the proposed method can capture the spatial distribution characteristics of strata through continuous sections.

Table 5

Losses of three reconstruction methods in geological body reconstruction task.

Index	IDW	Ordinary kriging	Proposed method
GA	0.8895	0.8746	**0.9329**
MSE	855.0563	1027.392	**476.0925**
SSIM	0.812	0.8012	**0.9083**
MS-SSIM	0.7411	0.7118	**0.8768**

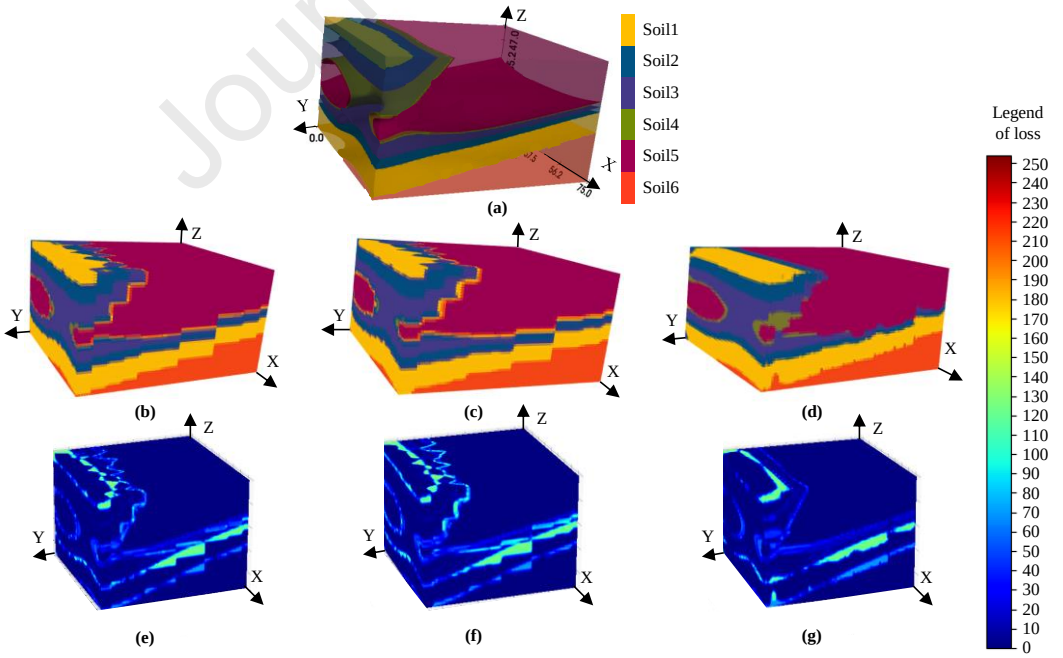


Fig. 13. Geological body reconstruction effects of three reconstruction methods. (a) reference strata, (b) strata reconstructed by IDW, (c) strata reconstructed by ordinary kriging, (d) strata reconstructed by the proposed method, (e) loss map of IDW, (f) loss map of ordinary kriging, and (g) loss map of the proposed method.

5. Application

In this section, the proposed method is applied to a practical engineering case involving complex strata. The Beijing East Sixth Ring Road (Jingha Expressway to Luyuan North Street) reconstruction project is located in the Tongzhou District of Beijing. It covers a total length of 16 kilometers. The twin tunnels section of the project is 9.2 kilometers long and features a maximum overburden of 59 meters. The outer diameter of the tunnel is 15.4 meters. The strata are composed of fill, silty clay, silt, fine sand, and medium sand. The tunnel was excavated through strata consisting of silty fine sand, fine sand, and silty clay. The stratigraphic distribution between K10 + 060 and K10 + 270 is shown in Fig. 14. Compared to the virtual stratum, the stratum distribution is complex and exhibits many interbedded phenomena.

The geological strata within the range of K10+060 to K10+270 were selected for simulation because the geological survey provided a geological profile at K10+065. To investigate the geology of the twin tunnels project, three rows of boreholes were drilled, as shown in Fig. 15.

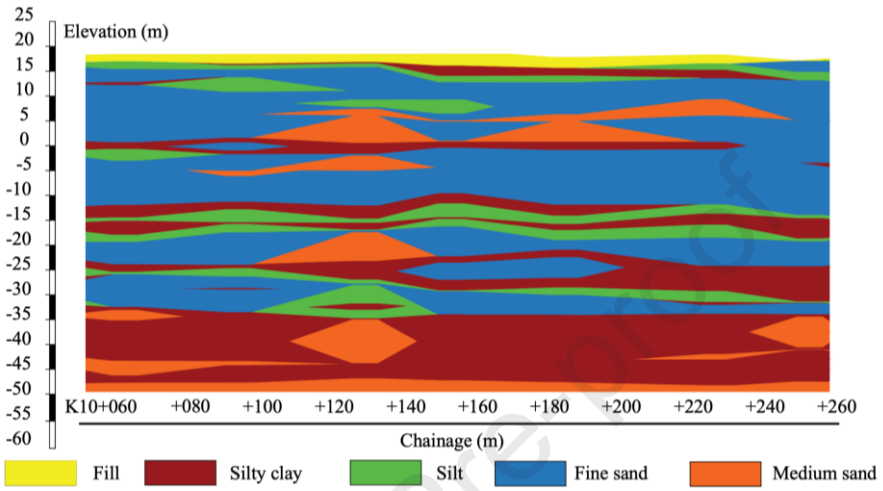


Fig. 14. Engineering geological profile.

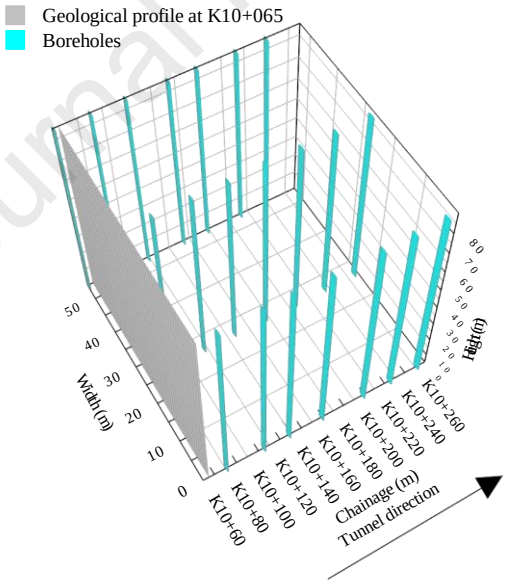


Fig. 15. Arrangement of boreholes.

The geological survey data from these boreholes were combined and transformed into contaminated images, which were used to create cross-sectional reconstructions. Through repeated modeling, the most likely formation distribution and the associated uncertainties are shown in Fig. 16. It can be found that, as in Section 4.2, there will be greater uncertainty at the location where the formation changes greatly (including the change of formation thickness and formation elevation), and the greatest uncertainty is located on the boundary of the formation.

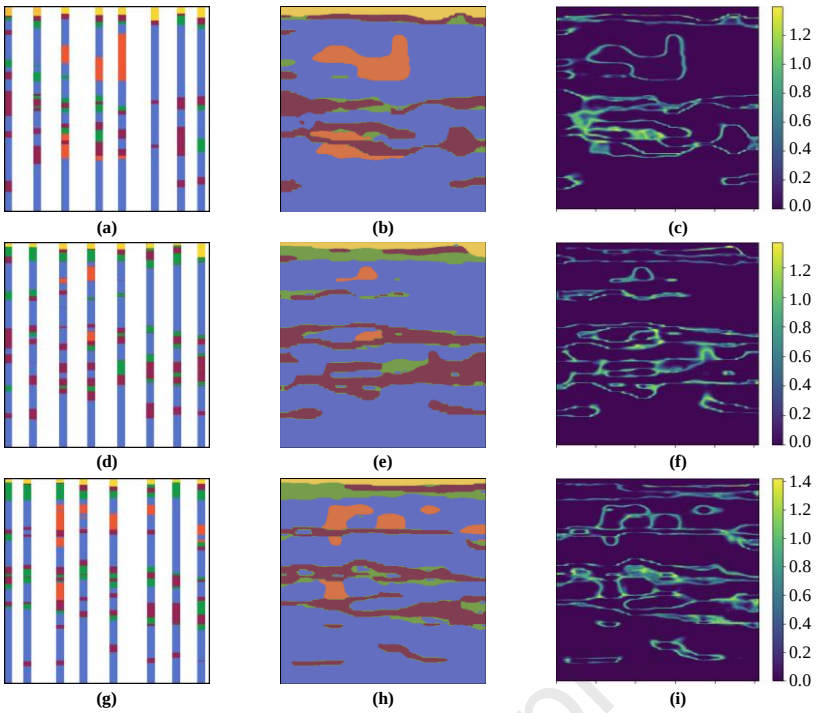


Fig. 16. Input data and reconstruction results for three sections of the reconstruction task. (a) left line borehole data, (b) most likely formation distribution of left line, (c) uncertainties of left line, (d) midline line borehole data, (e) most likely formation distribution of midline line, (f) uncertainties of midline line, (g) right line borehole data, (h) most likely formation distribution of right line, and (i) uncertainties of right line.

Accurate geological data is often unavailable in practical engineering projects. To evaluate the model's effectiveness in complex geological conditions, borehole data is used as a reference for assessing reconstruction results. Seven boreholes in a section are used for reconstruction, and the results are compared at corresponding positions with the stratigraphic distribution in the last borehole. The boreholes are numbered from left to right, as shown in Fig. 16. The GA results of stratigraphic distribution verified by different boreholes are shown in Fig. 17. As shown in Fig. 17, the global accuracy of the model, trained exclusively on virtual data, ranges from 0.5 to 0.9 for complex engineering geology. The lower GA value is attributed to the fact that the formation type and distribution in the corresponding borehole differ significantly from those in the surrounding boreholes. The boreholes with lower GA values have a greater impact on the accurate reconstruction of the geological model.

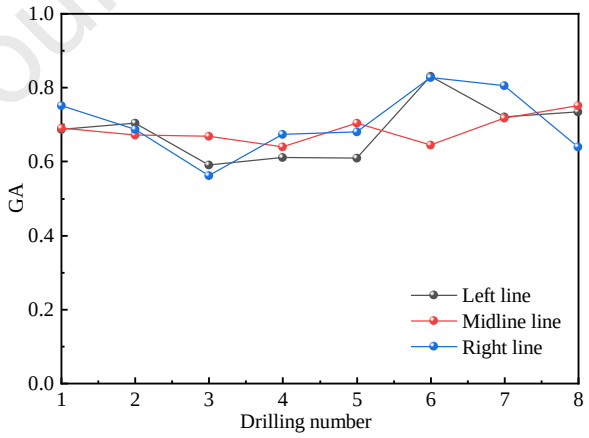


Fig. 17. Section reconstruction effect.

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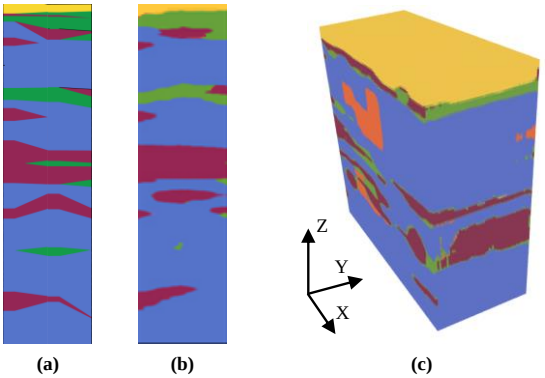

Fig. 18. Geological body reconstruction results. (a) geological profile at K10+065 provided by geological surveying, (b) geological profile at K10+065 obtained from reconstructed voxel model, and (c) reconstructed voxel model.

The reconstructed sections, along with their locations, were fed into the interpolation model, resulting in the reconstructed voxel model as shown in Fig. 18c. The geological strata distribution (Fig. 18b) for the section at K10+065 was obtained from the voxel model and compared with the data provided by the geological surveying unit (Fig. 18a). The results demonstrate that the proposed method, trained on virtual data, can rapidly reconstruct potential stratigraphic distribution patterns in complex urban geological settings.

6. Discussion

In this section, the influence of neural network model architecture, reconstruction framework, input data, and potential limitations of the proposed method were discussed.

6.1. Effect of neural network model architecture on 3D geological body reconstruction performance

In order to improve the reconstruction effect, an interpolation model including a skip connection and a self-attention mechanism is proposed. The architecture is shown in Fig. 4. In order to verify the validity of the modification, this study established two comparison models, U-Net and Context Encoder. The U-Net excludes the self-attention layer, while the Context Encoder eliminates the skip connection from the encoder to the decoder on the basis of the U-Net. The three models were trained on the same database described in Section 3.1 and tested on the simulated geological body described in Section 4.1. The results achieved on the test data are shown in Table 6. The reconstruction performance is shown in Fig. 19. The modified network demonstrates superior performance across all four evaluation metrics. The U-Net performs better than the Context Encoder. Therefore, the skip connection and self-attention mechanism can better capture the strata distribution characteristics of adjacent sections to reconstruct the intermediate section.

Table 6
Quantitative performance of three model architectures on test data (optimal value for each metric highlighted in bold).

Index	U-Net	Context Encoder	U-Net-SENet
GA	0.9261	0.926	**0.9329**
MSE	533.0941	539.3661	**476.0925**
SSIM	0.9013	0.9012	**0.9083**
MS-SSIM	0.8656	0.863	**0.8768**

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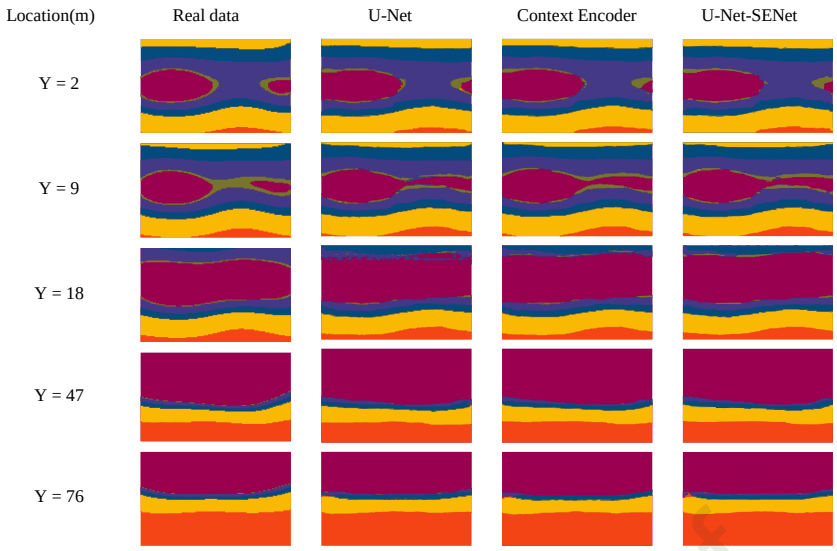

Fig. 19. Performance of three model architectures in reconstructing geological bodies on selected sections.

6.2. Effect of reconstruction framework on reconstruction performance and efficiency

In contrast to the framework introduced in Section 2.1, Shi and Wang (2022) developed a sequential reconstruction framework based on non-parallel cross-sections. This approach ensures that each reconstructed cross-section incorporates the maximum amount of borehole information possible. The framework assumes that the generated cross-sections are accurate. Virtual boreholes are extracted from these sections to provide accurate borehole information for the subsequently generated sections. The generation of cross-sections should follow the principles mentioned below:

$$\begin{cases} \text{X}, & if \ \arg\max DP_{\text{X}} > \arg\max DP_{\text{Y}} \\ \text{Y}, & if \ \arg\max DP_{\text{Y}} > \arg\max DP_{\text{X}} \\ \text{X } or \text{ Y}, & if \ \arg\max DP_{\text{X}} = \arg\max DP_{\text{Y}} \end{cases} \tag{32}$$

where X and Y represent the orientation of the section to be generated, DP_{X} represents the number of boreholes on the X-direction section to be generated, and DP_{Y} represents the number of boreholes on the Y-direction section to be generated. If the values of DP_{X} and DP_{Y} are equal, the direction is determined randomly.

Based on the reconstruction process and assumptions of Shi and Wang (2022), a new reconstruction framework based on non-parallel sections can be established, as shown in Fig. 20. This reconstruction framework can be divided into three parts: data acquisition, reconstruction decision-making, and cross-section reconstruction. Since it is similar to the framework in Section 2.1, only the differences between the two frameworks are introduced here.

Reconstruction decision-making. Since the section reconstruction is only based on the repair task, only the direction and location of the reconstructed section are determined at this stage. First, four boreholes are aligned along the X-axis, while in the Y-axis direction, at most, two boreholes are aligned. This means there is more information and lower uncertainty in the reconstruction of cross-sections along the X-axis. Therefore, the modeling of cross-sections along the X-axis should be given priority. Once we have accurate reconstruction results, virtual boreholes are generated from these results (depicted as red dashed circles). Then, a maximum of three boreholes is aligned along the Y-axis, but there are no boreholes aligned along the X-axis that can form new cross-sections. Therefore, the modeling of cross-sections along the Y-axis should be reconstructed. If the number of boreholes aligned along the X-axis and Y-axis is consistent. The cross-sections parallel to the X-axis or Y-axis are reconstructed randomly. This iterative process continues until the desired level of accuracy is achieved, indicating the completion of the reconstruction.

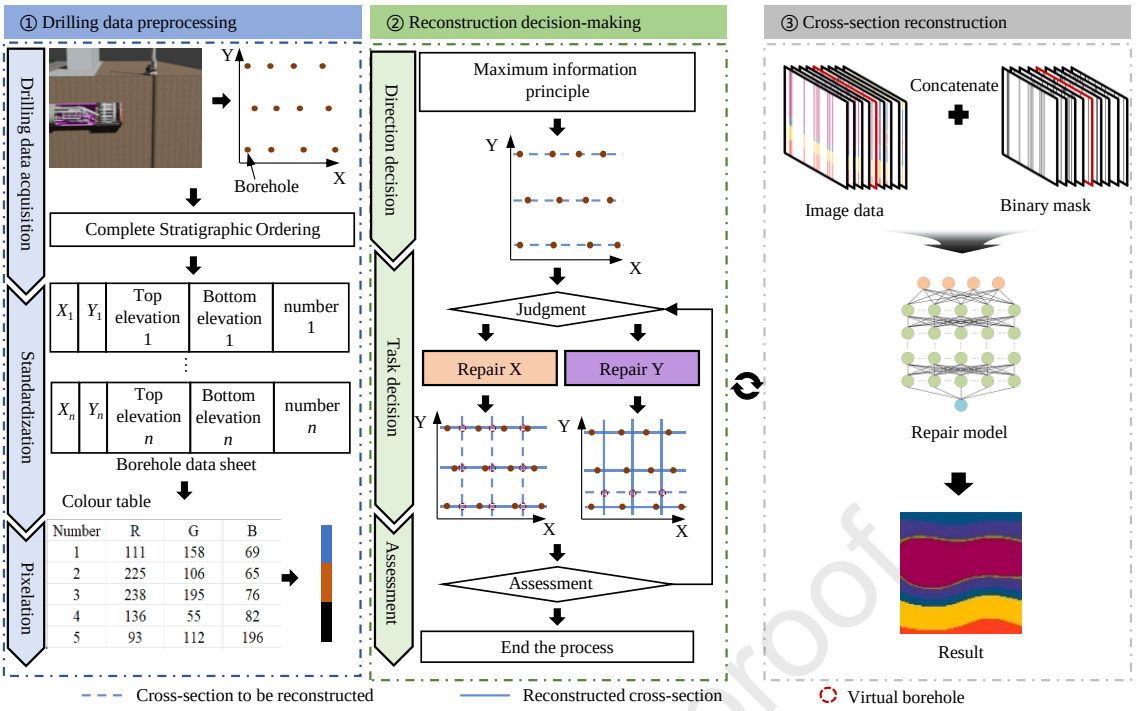

Fig. 20. Reconstruction process of the non-parallel cross-section reconstruction framework.

In order to verify the influence of the two frameworks on the reconstruction efficiency, two frameworks are used for geological reconstruction tasks of various sizes. The time recording starts after loading the machine learning model (as the initial loading of the trained machine learning model takes a relatively longer time) and stops until the reconstruction tasks are completed. The running time of the two frameworks at different scales is shown in Fig. 21. The parallel reconstruction framework reconstructs a middle section using adjacent sections. The time to obtain the adjacent section data is negligible, and the main time consumption is in the section reconstruction. The time consumption of the parallel reconstruction framework is linearly related to the number of sections reconstructed. In contrast, the non-parallel reconstruction framework needs to extract data from the reconstructed sections to guide the reconstruction of new sections. The number of extracted boreholes is positively correlated with the number of sections reconstructed. Here, the main time consumption is in extracting boreholes. Additionally, the time consumption for each section is also positively correlated with the total number of sections reconstructed. Thus, the total time consumption of the non-parallel reconstruction framework is proportional to the square of the number of sections reconstructed. The computational load of the two frameworks can be shown as follows:

$$\begin{cases} Cost_{\mathrm{p}} = s \cdot t_1 + a_1 \\ Cost_{\mathrm{n}} = s^2 \cdot t_2 + a_2 \end{cases} \tag{33}$$

where $Cost_{\mathrm{p}}$ represents the computational load of the parallel reconstruction framework, $Cost_{\mathrm{n}}$ represents the computational load of the non-parallel cross-section reconstruction framework, t_1 and t_2 represent the unit time required to reconstruct one section of the two frameworks, and s represents the dimension of the reconstruction area. Additionally, a_1 and a_2 show the computational cost caused by other factors of the two frameworks. When the s reaches 480, the non-parallel reconstruction framework takes 70 times more time than its parallel counterpart. This demonstrates that the parallel reconstruction framework outperforms the non-parallel reconstruction framework in terms of computational cost.

The reconstruction results of the two frameworks are compared, and the methods described in Section 3.3 are used to evaluate the results. The reconstruction results of five randomly selected interpolated surfaces (surfaces other than the initial borehole locations) are shown in Table 7, and the difference cloud diagrams are shown in Fig. 22. The parallel reconstruction framework has a lower MSE error compared to the non-parallel reconstruction method, and it exhibits a higher structural similarity with the real images. This is caused by the assumptions made in the non-parallel reconstruction frameworks. In the non-parallel reconstruction framework, virtual borehole data is extracted from the reconstructed sections, and this data is directly utilized from one direction in another direction (as shown in Fig. 23), assuming that the spatial distribution of these boreholes in both directions is consistent, which does not align with the actual scenario. Therefore, the parallel reconstruction framework exhibits better adaptability in the reconstruction method proposed in this study.

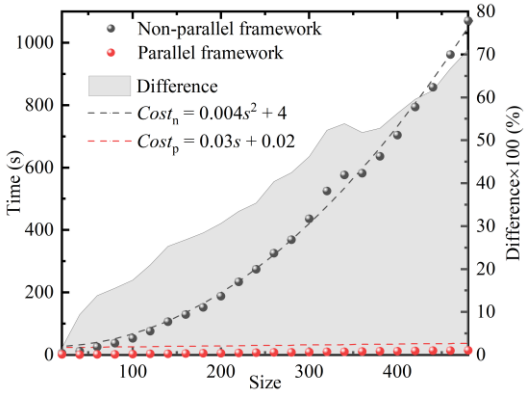

Fig. 21. Computational cost of two reconstruction frameworks for tasks of various sizes.

Table 7

Quantitative description of reconstruction performance of two reconstruction frameworks (optimal value for each metric is highlighted in bold).

Index	Non-parallel framework	Parallel framework
GA	0.9064	**0.9329**
MSE	709.9515	**476.0925**
SSIM	0.8768	**0.9083**
MS-SSIM	0.8233	**0.8768**

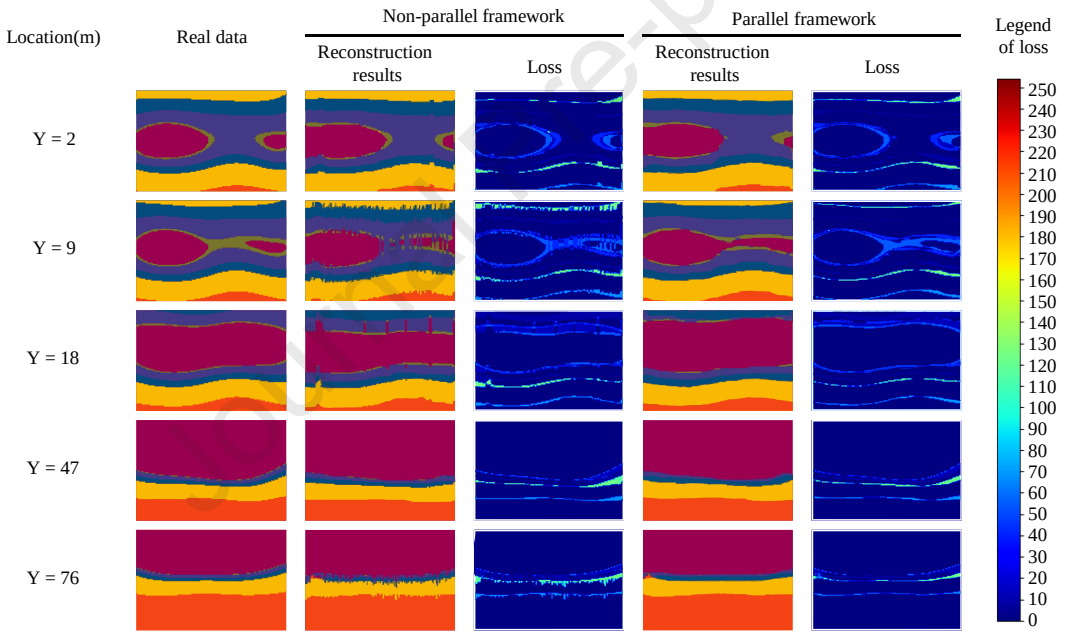

Fig. 22. Performance of two reconstruction frameworks in reconstructing geological bodies on partial sections.

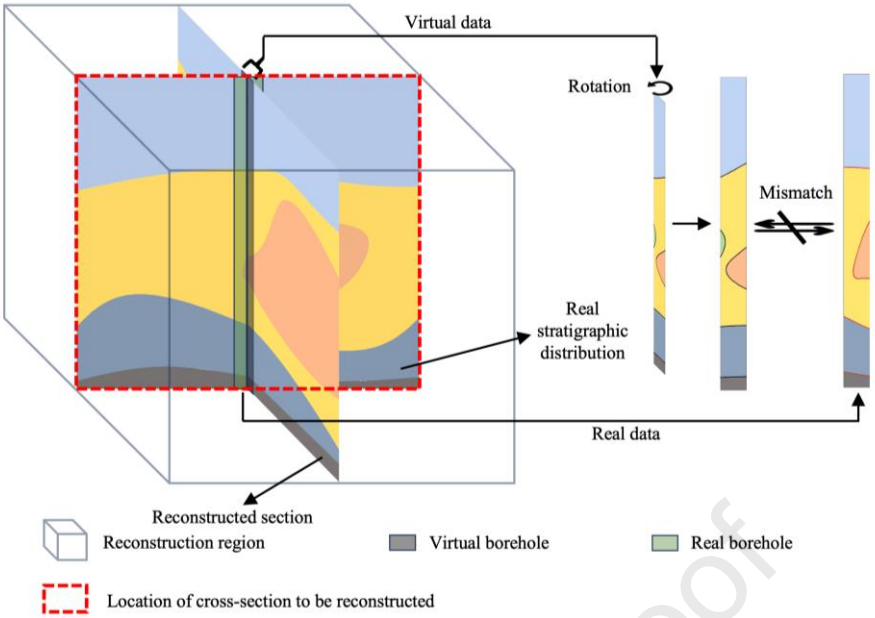

Fig. 23. Geological distribution of virtual boreholes generated during the reconstruction process in a non-parallel reconstruction framework may not necessarily match in both directions.

6.3. Effect of reconstruction framework on reconstruction performance and efficiency

The quality of 3D reconstruction depends on the arrangement and number of boreholes. To test the performance of the proposed method under different conditions, two sets of 512 × 512-pixel reconstruction scenes are built. In both scenes, the number of boreholes ranged from 2 to 40, with each borehole having a width of 4 pixels. In the first scene, the boreholes are evenly spaced, while in the second, the boreholes are randomly distributed. The information coverage (IC) is used to measure the difficulty of the reconstruction task, which represents the proportion of known information in the reconstruction task to the overall information. In the case of image reconstruction, this refers to the proportion of known pixels to the total number of pixels.

$$IC = \frac{n_b \times b_w \times b_h}{I} \times 100\% \qquad (34)$$

where n_b represents the number of boreholes, b_w represents the number of pixels corresponding to the width of a borehole in the image, b_h represents the number of pixels corresponding to the height of a borehole in the image, and I represents the total number of pixels in the image. In this paper, b_w is 4, b_h is 512, and I is 512 × 512. Fig. 24 shows that the arrangement of equally spaced boreholes can achieve more accurate reconstruction results with the same number of boreholes.

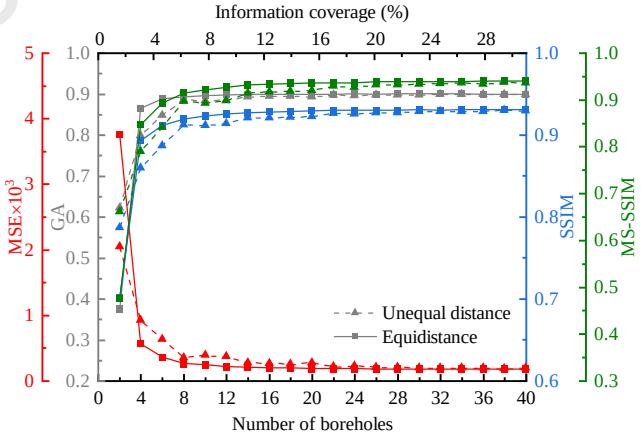

Fig. 24. Reconstruction effects of different arrangements and numbers of boreholes.

When there are two boreholes, the non-equidistant distribution achieves better reconstruction performance. This is because the distance between the two boreholes in an equidistant distribution is too large. This distance exceeds the receptive field of the convolutional network. It is difficult for the model to capture the distribution characteristics of the stratum. When the two boreholes are randomly distributed, the probability of the distance between them being smaller is higher. The stratigraphic distribution information in the borehole can be captured for the reconstruction of some areas. When boreholes are equally spaced, four boreholes can yield relatively accurate reconstruction results,

while eight boreholes can yield more accurate reconstruction results. In contrast, a completely random arrangement of boreholes requires more boreholes to achieve reconstruction performance similar to that of equally spaced boreholes.

6.4. Limitation

The proposed parallel cross-section-based reconstruction framework effectively captures the spatial distribution characteristics of strata and reconstructs potential geological models quickly and conveniently. The framework can also measure stratum uncertainty through some methods. However, it is important to acknowledge certain limitations that should be clearly explained in its practical application.

(1) It should be noted that the proposed framework is a reconstruction method based on the borehole data. When a stratum is not exposed in the borehole data, it will lead to the neglect of the stratum information in the reconstruction results.

(2) Although this method can quickly reconstruct a possible stratigraphic distribution state to assist field personnel in decision-making, the method is a linear process, and the reconstruction error will accumulate and amplify. But we can reduce the reconstruction error by repeated modeling. In practical use, the demand for reconstruction should be clarified, and rapid reconstruction or repeated reconstruction (to obtain the most likely stratigraphic distribution) should be performed according to requirements.

(3) The proposed method is a supervised learning method, so the database has a significant impact on the accuracy of the reconstruction results. The model can only reconstruct the strata features that have appeared in the training data, so the database should contain as many diverse geological features as possible. Section 5 demonstrates that the model can be trained on virtual data and adapted for real engineering applications. When the amount of engineering data increases, the geological profile data drawn by experienced engineers can be added to the data set to enhance the generalization ability of the model.

7. Conclusions

A novel 3D geological body reconstruction method was proposed. The proposed method comprises a repair model used to reconstruct cross-sections, an interpolation model used to reconstruct geological bodies, and a parallel cross-section reconstruction framework. The proposed method can significantly reduce the computational cost while maintaining the reconstruction effect. The following conclusions could be drawn:

(1) The proposed method harnesses neural networks' feature-extraction capabilities to capture global contextual information from borehole data, facilitating reconstruction of strata's distribution. The GA achieved 0.94 in the cross-section reconstruction task and 0.93 in the 3D geological model reconstruction.

(2) The proposed method uses dropout to control the knowledge learned by the model. Combined with repeated modeling experiments, it can evaluate the uncertainty of reconstructed strata and generate the most likely cross-sectional stratigraphic distribution by statistically analyzing repeated reconstruction results.

(3) The proposed method achieves linear time complexity by reformulating geological model reconstruction as the parallel reconstruction of individual cross-sections. Compared with a non-parallel framework, it reduces processing time by up to 98% when reconstructing 400 cross-sections. Leveraging borehole data, the model facilitates rapid 3D geological modeling of urban subsurface environments. Owing to its rapid modeling capability and through coordinate transformation, the method can be extended to model mountainous terrain using Measurement While Drilling data, thus supporting drill-and-blast operations.

(4) Distribution and number of boreholes can affect the reconstruction results. Due to the limitations of the receptive field in the convolutional network, the proposed method has difficulty in establishing the connections between the boreholes when their spacing is too large (i.e. exceeding twice the receptive field, which corresponds to 3 or fewer boreholes in this study), leading to degraded reconstruction performance. When the borehole spacing meets the receptive field requirements, equidistantly distributed boreholes yield better reconstruction results.

The proposed method incorporates the IDW approach as prior knowledge into the deep model, achieving high reconstruction accuracy on validation data. However, like most interpolation techniques, it relies on borehole data and cannot infer unexposed features (e.g. boulders). In future work, we will integrate multisource data, such as elastic and seismic wave measurements, with the reconstructed model to enhance its accuracy and robustness.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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List of symbols

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x_r	Real data
x_d	Dirty data generated by x_r
x_g	Output of the generator
E	Expected value
G	Generator network
D	Discriminator network
L_{adv}	Adversarial loss used for training discriminator
L_1	Mean absolute error loss
L_{GDL}	Gradient difference loss
L_{NPCL}	Normalized product correlation loss
H_{image}	Height of image
W_{image}	Width of image
λ	Anisotropy ratio

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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